

The following article is from the NOS, a Dutch news outlet, and was published on the 27th of November 2025.

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Jongeren herkennen online oplichting minder goed dan ze zelf denken

Vier op de tien Nederlanders hebben een te hoge verwachting van zichzelf bij het herkennen van online oplichting. Ze denken nep en echt beter te kunnen onderscheiden dan in werkelijkheid zo is. Dat blijkt uit [onderzoek](#)  in opdracht van het ministerie van Justitie en Veiligheid.

Verder blijkt uit het onderzoek dat mensen onder de 34 jaar gevoeliger zijn voor online oplichting dan mensen die ouder zijn. Terwijl mensen jonger dan 34 jaar wel het vaakst denken online misleiding goed te kunnen herkennen.

De achttienhonderd deelnemers kregen voor het onderzoek een e-mail, sms, WhatsApp-bericht, website en een videobericht voorgelegd. De ene helft kreeg de echte variant te zien, bij de andere helft waren er signalen van misleiding toegevoegd. Een van de voorbeelden was een korte video van illusionist Victor Mids, met daarbij een deepfake-variant waaraan met behulp van AI een namaak-boodschap was toegevoegd.

De illusionist vond het verbijsterend dat de mensen daarin traptten, hoewel hij beroepshalve weet dat mensen gemakkelijk te misleiden zijn. "Je brein maakt ergens in je besluitvorming een verkeerde aanname, en voor je het weet heb je bijvoorbeeld geklikt op een foutieve link."

Zelfoverschatting

Het ministerie en de onderzoekers willen de voorbeelden die ze in het onderzoek gebruikten niet naar buiten brengen, om online oplichters niet op ideeën te brengen. Die worden steeds slimmer en geraffineerder, stellen ze.

Bij de beoordeling tussen nep en echt letten de onderzochte personen vooral op taalgebruik, de afzender en onlogische verzoeken. Een groot gedeelte van de Nederlanders verwacht dat online oplichting alleen maar vaker zal voorkomen en dat die steeds moeilijker te herkennen zal zijn.

Met het onderzoek 'Laat je niet Interneppen' wil de overheid burgers [waarschuwen](#) . "Doordat het verschil tussen echt en n t echt zo klein is, en veel mensen zichzelf overschatten, lopen we meer risico dan we vaak denken. En juist daarom kan het iedereen overkomen", zegt demissionair minister van Justitie en Veiligheid Van Oosten.

Tips

Yianne Spoormans, cybercrime-specialist bij de politie, heeft wel wat tips om de steeds geraffineerdere aanpak van online oplichters door te prikken.

"Ze proberen je te overtuigen snel te handelen, bijvoorbeeld om een noodsituatie af te wenden, of omdat een product bijna uitverkocht is. Door deze tijdsdruk kun je niet meer kritisch nadenken", legt zij uit. "Dus moet er per se snel gehandeld worden? Wees dan alert en doe een stap terug. Bouw tijd in om de feiten en de afzender te controleren."

Ook kan het helpen of iemand die je vertrouwt er even naar te laten kijken of om advies te vragen. Spoormans: "Klopt het niet of twijfel je? Klik dan weg."

The English translation of the article, created with ChatGPT; The following article is from the NOS, a Dutch news outlet, and was published on the 27th of November 2025.

Young people are less able to recognise online scams than they think

Four in ten Dutch people overestimate their ability to recognise online scams. They think they are better at distinguishing between fake and genuine content than they actually are. This is shown by research commissioned by the Dutch Ministry of Justice and Security.

The research also shows that people under the age of 34 are more vulnerable to online scams than older people. At the same time, people under 34 are the most likely to believe that they are good at recognising online deception.

For the study, the 1,800 participants were shown an email, text message, WhatsApp message, website and video message. Half of the participants were shown the genuine version, while the other half were shown versions containing signs of deception. One of the examples was a short video featuring illusionist Victor Mids, together with a deepfake version in which a fake message had been added using AI.

The illusionist found it astonishing that people fell for it, although his profession has taught him that people can easily be deceived.

“At some point in your decision-making process, your brain makes an incorrect assumption, and before you know it, you may have clicked on a fraudulent link.”

Overconfidence

The Ministry and the researchers do not want to publicly release the examples used in the study, because they do not want to give online scammers new ideas. According to them, scammers are becoming increasingly clever and sophisticated.

When deciding whether something was fake or genuine, participants mainly paid attention to the language used, the sender, and requests that seemed illogical. A large proportion of Dutch people expect online scams to become more common and increasingly difficult to recognise.

With the study “*Laat je niet Interneppen*”, the government aims to warn citizens about online scams.

“Because the difference between genuine and almost genuine content is so small, and because many people overestimate their own abilities, we are at greater risk than we often realise. That is exactly why it can happen to anyone,” says Van Oosten, the outgoing Minister of Justice and Security.

Tips

Yianne Spoormans, a cybercrime specialist with the police, offers several tips for recognising the increasingly sophisticated methods used by online scammers.

“They try to convince you to act quickly, for example to prevent an emergency or because a product is almost sold out. Because of this time pressure, you are no longer able to think critically,” she explains. “So, if you are being told that you have to act immediately, be alert and take a moment before responding. Give yourself time to check the facts and verify the sender.”

It can also help to ask someone you trust to look at the message or to ask them for advice. Spoormans concludes:

“If something does not seem right or you are unsure, close the screen.”

The challenge

The article illustrates how many people, in particular those below the age of 34, have the assumption that they are able to adequately distinguish real messages from fraudulent ones. Through an experiment it was shown that it was more challenging for the participants to make the distinction than what they had anticipated beforehand. This visualizes that online fraud is becoming more of a problem for everyone of all ages, not only elderly people as is often assumed.

Online fraud is a modern societal challenge that can impact many people, having a broad relevance for both individuals and organizations. Particularly in this day and age where technological developments, like AI, make it increasingly difficult to differentiate between real messages and fake ones.

Scientific articles

1. Can educational interventions reduce susceptibility to financial fraud? .
(Original research paper)

Burke, J., Kieffer, C., Mottola, G., & Perez-Arce, F. (2022). Can educational interventions reduce susceptibility to financial fraud? *Journal of Economic Behavior & Organization*, 198, 250–266. <https://doi.org/10.1016/j.jebo.2022.03.028>

The article is relevant because it investigates whether education can help people recognise and avoid fraud. The researchers found that a short educational intervention can help with preventing and recognising online fraud. This connects to the NOS article because people think that they are often good at recognising them. The study suggests that better education and awareness could help people recognise scams more effectively and protect themselves from becoming victims.

2. The psychology of internet fraud victimisation (review)

Norris, G., Brookes, A., & Dowell, D. (2019). The psychology of internet fraud victimisation: A systematic review. *Journal of Police and Criminal Psychology*, 34, 231–245. <https://doi.org/10.1007/s11896-019-09334-5>

This article is relevant, because it investigates why people are vulnerable to online fraud. The review shows that vulnerability to online fraud is influenced by different psychological factors, like the characteristics of the fraudulent message, previous experiences and individual characteristics. All these factors help us understand why sometimes people fail to recognise online scams, even when they believe they are able to do so.

3. Navigating online perils: Socioeconomic status, online activity lifestyles, and online fraud targeting and victimization of old adults in China (research paper)

Wang, D., Duan, Y., & Jin, Y. (2025). "Navigating online perils: Socioeconomic status, online activity lifestyles, and online fraud targeting and victimization of old adults in China." *Computers in Human Behavior*, 162, 108458. <https://doi.org/10.1016/j.chb.2024.108458>

Within this article, the relationship between socioeconomic status, online behaviour and the risk of online fraud is researched. The results showed that older adults (aged 50 and above in China) with a higher socioeconomic status were more likely to be targeted by online fraud. The study also defined four different online activity lifestyles, basic, entertainment, risk and advanced. Which were associated with different levels of online fraud targeting. This article is relevant to our challenge, as it demonstrates that socioeconomic characteristics and online behaviour can influence exposure and vulnerability to online fraud.

The ERD

What should the database be able to answer?

- We want to create a database to see who is most likely to become a victim of online fraud and how the reporting differentiates

The entities, their attributes with PKs and FKs

- Person
 - Person_ID (PK)
 - Age
 - Occupation
 - Income
 - Education_level
 - Gender
- Fraud Incident
 - Fraud_Incident_ID (PK)
 - Fraud_type_ID (FK)
 - Person_ID (FK)
 - Channel_ID(FK)
 - Incident_date
 - Financial_loss_amount
- Report
 - Report_ID (PK)
 - Fraud_incident_ID (FK)
 - Organisation_ID(FK)
 - Report_date
- Fraud Type
 - Fraud_type_ID (PK)
 - Fraud_kind
- Communication channel
 - Channel_ID (PK)
 - Channel_kind
- Organisation
 - Organisation_ID (PK)
 - Organisation_name
 - Organisation_type
- Consequences
 - Consequence_ID (PK)
 - Fraud_incident_ID (FK)
 - Emotional_problem
 - Financial_problem

How the data interacts

- A person can experience multiple fraud incidents, but each fraud incident belongs to one person.
- A fraud incident has one fraud type, but a fraud type can occur in multiple fraud incidents.
- A fraud incident uses one communication channel, but a communication channel can be used for multiple fraud incidents.
- A fraud incident can have multiple reports, but each report belongs to one fraud incident.
- An organisation can receive multiple reports, but each report is made to one organisation.
- A fraud incident can have multiple consequences, but each consequence belongs to one fraud incident.

The relationships

- Person → experiences → fraud incident: one-to-many
- Fraud type → occurs at → fraud incident: one-to-many
- Communication channel → is used for → fraud incident: one-to-many
- Fraud incident → generates → report: one-to-many
- Organisation → receives → report: one-to-many
- Fraud incident → has → consequences: one-to-one

The setup of the ERD was inspired by the data from Centraal Bureau Statistiek (CBS), the statistical agency of the Netherlands;

<https://www.cbs.nl/nl-nl/longread/rapportages/2025/online-veiligheid-en-criminaliteit-2024/4-online-oplichting-en-fraude>

Normalizing the ERD

1NF: Atomic Values

The first database normalization step is First Normal Form (1NF). In this step, each cell should only contain one atomic value, meaning that multiple values within one cell are not allowed. To achieve this, the attributes with multiple values can be separated into individual rows in related tables. The following example shows a 1NF violation:

Before 1NF, with example data:

Fraud_incident_ID	Person_ID	Fraud_type	Communication_channel	Report_date
FI1	P1	Phishing	Email, Phone	04-07-2022
FI2	P2	Phishing, Identity theft	Social Media	16-02-2024
FI3	P3	Investment scam	Email	23-10-2025

In the table above two cells contain more than one value. For FI1, the attribute Communication_channel contains both Email and Phone. FI2 also has an 1NF violation, as Fraud_type consists of Phishing and Identity Theft. Therefore, the table does not satisfy 1NF as the values are not atomic. To solve this problem, the values could be separated into individual rows in relationship tables.

After 1NF, with example data:

Fraud type table:

Fraud_incident_ID	Fraud_type
FI1	Phishing
FI2	Phishing
FI2	Identity Theft
FI3	Investment Scam

Communication channel table:

Fraud_incident_ID	Communication_channel
FI1	Email
FI1	Phone
FI2	Social Media
FI3	Email

After these adaptations, each cell contains only one value, meaning that the attributes are atomic and the tables meet the 1NF requirements.

2NF: No Partial Dependencies

In our Report table, we need two columns together to know exactly which row we're looking at: Fraud_incident and Organisation_ID. That is because one incident can be reported to several organisations, so neither column alone is enough to point to one specific row, you need both.

Now imagine Organisation_name and Organisation_type was also written directly into the Report table. Those two things do not care which incident it was, they only belong to the organisation. So every time the same organisation showed up in a new report, we'd be typing its name in again, even though nothing about the organisation itself had changed. So if an organisation's name ever changed, we'd have to go find and fix it in every single row it appeared in.

Before, with example data:

Fraud_incident_ID	Organisation_ID	Organisation_name	Organisation_type	Report_date
1	O1	Police Netherlands	Police	2026-01-11
2	O1	Police Netherlands	Police	2026-01-16
1	O2	ABN AMRO	Bank	2026-01-11

The problem we want to avoid in the example above, is the repeated information. In this case, the Police Netherlands/Police that appears twice.

After, with example data:

Organisation table:

Organisation_ID	Organisation_name	Organisation_type
O1	Police Netherlands	Police
O2	ABN AMRO	Bank

Report Table:

Report_ID	Fraud_incident_ID	Organisation_ID	Report_date
R1	1	O1	2026-01-11
R2	2	O1	2026-01-16
R3	1	O2	2026-01-11

Now each organisation's name and type is written down just once, in its own table. The Report table just points to it using Organisation_ID, instead of repeating the details every time.

3NF: No Transitive Dependencies

The third step of database normalization is Third Normal Form (3NF). A database is in 3NF when it already meets the requirements of 1NF and 2NF, and when non-key attributes do not depend on other non-key attributes. Instead, each non-key attribute should depend directly on the primary key.

In our database, a transitive dependency could occur if information about the fraud type is stored directly in the Fraud_Incident table. For example, the table could contain both *Fraud_type_ID* and *Fraud_kind*. The problem is that *Fraud_kind* does not depend directly on

Fraud_incident_ID. Instead, it depends on *Fraud_type_ID*, because each fraud type has one corresponding fraud kind.

This can be shown as:

Fraud_incident_ID → Fraud_type_ID → Fraud_kind

For example, if multiple fraud incidents are classified as phishing, the value "Phishing" would be repeated for every incident. This creates unnecessary duplicate data and could lead to inconsistencies. If information about the fraud type needed to be changed, it would need to be changed in multiple rows.

Before 3NF:

Fraud_incident_ID	Person_ID	Fraud_type_ID	Fraud_kind
FI1	P1	FT1	Phishing
FI2	P2	FT1	Phishing
FI3	P3	FT2	Identity Theft
FI4	P1	FT1	Phishing

The problem is that *Fraud_kind* depends on *Fraud_type_ID* instead of directly on *Fraud_incident_ID*. The value "Phishing" is also repeated multiple times.

After 3NF:

To solve this problem, we created a separate *Fraud_Type* table.

Fraud_Type

Fraud_type_ID	Fraud_kind
FT1	Phishing
FT2	Identity Theft

Fraud_Incident

Fraud_incident_ID	Person_ID	Fraud_type_ID
FI1	P1	FT1
FI2	P2	FT1
FI3	P3	FT2
FI4	P1	FT1

After this change, each fraud type is stored only once in the *Fraud_Type* table. The *Fraud_Incident* table only stores the *Fraud_type_ID* as a foreign key to link each incident to its type.

This removes the transitive dependency and reduces duplicated data. It also makes the database easier to maintain, because information about a fraud type only needs to be updated once. Therefore, our database satisfies the requirements of Third Normal Form (3NF).

Normalization of the ERD

1NF: Atomic Values

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After 1NF, with example data:

Fraud type table:

Fraud_incident_ID	Fraud_type
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FI2	Phishing
FI2	Identity Theft
FI3	Investment Scam

Communication channel table:

Fraud_incident_ID	Communication_channel
FI1	Email
FI1	Phone
FI2	Social Media
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After these adaptations, each cell contains only one value, meaning that the attributes are atomic and the tables meet the 1NF requirements.

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Organisation table:

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This can be shown as:

Fraud_incident_ID → Fraud_type_ID → Fraud_kind

For example, if multiple fraud incidents are classified as phishing, the value "Phishing" would be repeated for every incident. This creates unnecessary duplicate data and could lead to inconsistencies. If information about the fraud type needed to be changed, it would need to be changed in multiple rows.

Before 3NF:

Fraud_incident_ID	Person_ID	Fraud_type_ID	Fraud_kind
FI1	P1	FT1	Phishing
FI2	P2	FT1	Phishing
FI3	P3	FT2	Identity Theft
FI4	P1	FT1	Phishing

The problem is that *Fraud_kind* depends on *Fraud_type_ID* instead of directly on *Fraud_incident_ID*. The value "Phishing" is also repeated multiple times.

After 3NF:

To solve this problem, we created a separate *Fraud_Type* table.

Fraud_Type

Fraud_type_ID	Fraud_kind
FT1	Phishing
FT2	Identity Theft

Fraud_Incident

Fraud_incident_ID	Person_ID	Fraud_type_ID
FI1	P1	FT1
FI2	P2	FT1
FI3	P3	FT2
FI4	P1	FT1

After this change, each fraud type is stored only once in the *Fraud_Type* table. The *Fraud_Incident* table only stores the *Fraud_type_ID* as a foreign key to link each incident to its type.

This removes the transitive dependency and reduces duplicated data. It also makes the database easier to maintain, because information about a fraud type only needs to be updated once. Therefore, our database satisfies the requirements of Third Normal Form (3NF).

Data sources

The first data source that we selected and adapted data from is the Canadian Anti-Fraud Centre (CAFC) dataset called “Canadian Anti-Fraud Centre Reporting Data” ([Canadian Anti-Fraud Centre Fraud Reporting System Dataset - Open Government Portal](#)). It was published in July 2023 and last updated in October 2025. It has the licence Open Government Licence - Canada. From this source, we randomly selected 200 reports, where Complaint Type = “Victim” and there was exactly one victim per report. The selected CAFC data is used to populate our Fraud_type, Communication_Channel, Fraud_Incident, Report and Organisation tables

The second data source that we have chosen is the Consumer Sentinel Network Data Book 2024 from the Federal Trade Commission (FTC) in the United States ([Consumer Sentinel Network Data Book 2024 | Federal Trade Commission](#)). It was published in March 2025. It has no licence, because work from the US government is in the public domain, so it is free to use. From this source we used the number of reports per fraud type for 2022, 2023 and 2024 (Appendix B3), and the reports and losses per contact method and per age group (pages 12 and 13). We only kept the fraud categories, which gave 123 report counts and 16 loss statistics.

CAFC and FTC are not subsets of each other.

Integrating data into the database

For the CAFC data, the schema had to be updated to accommodate the way age was notated. In the data, age is collected as a range as opposed to an exact number. So the attribute Age was removed and replaced with the attribute Age_range. In addition, Financial_loss_amount in the Fraud_Incident table now allows NULL values to facilitate the data.

Alongside this, the CAFC data also required some data cleaning and transformation. There were no empty cells, instead missing values were represented as text, such as “Not Available” and “Unknown” in the gender column, and “Not Available”, “Unknown” and “Deceased” in age. These missing values were replaced with NULL in Age_range and Gender. Besides that, the financial loss data also needed to be transformed. The dollar sign and comma separators, for example \$1,000.00, were removed to store the values as numbers. The amounts were then converted from Canadian dollars to US dollars using exchange rates corresponding to the year of each record. CAFC also uses \$0.00 for identity fraud, personal information and phishing when the loss cannot be determined. These values were therefore changed to NULL, while \$0.00 for other fraud types was retained as 0, as it represents no reported financial loss.

The dates were already in ISO format (YYYY-MM-DD), the same as in the mock data, so no conversion was required. Furthermore we checked the CAFC data for duplicate records and found no duplicate report numbers in the data. Finally, the data contained every column twice, once in English and once in French. To maintain consistent naming conventions in our database, only the English versions were used.

The FTC data required the schema to create two new tables. The FTC only gives totals per year and not separate reports, so the data does not fit in Person, Fraud_Incident or Report.

We added Report_Count for the number of reports per fraud type and year, and Loss_Statistic for the losses per contact method or age group. We also made Fraud_kind and Channel_kind case-insensitive, so "Phone call" and "Phone Call" are not stored twice.

The FTC data also needed to be transformed. Missing values are not marked: if someone did not fill in their age, the report is just left out of that table. There are no full dates, only years, which we stored as Report_year. One report can be counted in more than one subcategory, so the subcategories together are higher than the category total. Some names have a symbol behind them, like "Eye Care*", which we removed. Amounts like "\$502M" and percentages like "25%" were changed to numbers. Lastly, the FTC names were changed to the names the CAFC data already uses, for example "Text" became "Text message".

Example queries week 3

We re-ran the three example queries from week 3 using the real CAFC and FTC data and checked whether the results were as expected.

Query 1 (organisations ranked by number of reports) works without changes.

Query 2 (average loss by age bracket and channel) needed to be updated. The CAFC data mainly contains age ranges instead of exact ages. We therefore changed the query to use Age when available and Age_range otherwise, with "Unknown" for missing ages.

Query 3 (repeat victims) works correctly with the same logic as before. The query returns no rows for the real data because every person in the sample has only one incident. This is a meaningful result rather than an error. The query was also confirmed to work using the mock data, where repeat victims were found.

Query 4 (Canada vs. the United States) was added to compare the fraud types in the two real datasets. We added Query 4 so that we could use the real data and get an actual result from the query, since Query 3 does not return any results with the real data. It ranks the fraud type in each country and shows the difference between their shares.

Normalization realworld data

With the CAFC and FTC data the database is still normalized up to 3NF, no further adjustments had to be made.

Results queries

Overall, the queries produce meaningful results after the necessary updates. Query 1 correctly reflects how reports are stored, Query 2 provides useful information about the average loss by age and channel. Query 3 correctly shows that there are no repeat victims in the current sample. Query 4 also gives a useful comparison between Canada and the US, showing that Identity Fraud is the largest category in both, while some other fraud types have different shares between the countries. The results show that Identity Fraud is the most common fraud type in both countries, accounting for 62.5% of CAFC reports and 58.7% of FTC reports.

Evaluation of Query Results

Query 1: Expected. All 200 CAFC reports are linked to the Canadian Anti-Fraud Centre. The FTC data is aggregated, so no individual reports are linked to it and it shows 0.

Query 2: Partly expected. Phone calls causing high losses for ages 50-69 fits what I assumed. The very high averages for "Internet" (30-49) and "Unknown" were less expected, but they are based on few incidents, so a few large losses dominate. Several 0.00 averages mean those incidents had no loss.

Query 3: Expected. Each report in the CAFC sample belongs to a different person, so there are no repeat victims. The dataset cannot show repeat victimisation over time.

Query 4: Partly expected. Identity Fraud is the largest type in both countries (62.5% CAFC, 58.7% FTC, calculated within the fraud types present in both datasets). The biggest differences are Merchandise fraud (higher in the US) and Investment fraud (higher in Canada). Because the CAFC data is a sample, small differences should not be over-interpreted.

Results

===== Query 1 =====

	Organisation_name	Organisation_type	reports_handled
0	Canadian Anti-Fraud Centre	Law Enforcement	200
1	FTC Consumer Sentinel Network	Government	0

===== Query 2 =====

	age_bracket	Channel_kind	incident_count	avg_loss	loss_rank
0	30-49	Internet	9	59901.74	1
1	30-49	Internet-social network	7	8411.80	2
2	30-49	Direct call	5	2031.77	3
3	50-69	Direct call	8	54625.00	1
4	50-69	Other/unknown	2	17945.00	2
5	50-69	Internet	8	12023.92	3
6	50-69	Internet-social network	1	70.20	4
7	50-69	Email	3	0.00	5
8	70+	Internet	1	15702.00	1
9	70+	Internet-social network	3	9104.50	2
10	70+	Door to door/in person	1	1112.00	3
11	70+	Direct call	1	0.00	4
12	70+	Email	1	0.00	4
13	Under 30	Internet	4	968.45	1
14	Under 30	Email	3	333.33	2
15	Under 30	Internet-social network	3	282.91	3
16	Under 30	Direct call	2	0.00	4
17	Unknown	Email	7	106579.61	1
18	Unknown	Text message	3	73595.57	2
19	Unknown	Internet-social network	7	20911.00	3
20	Unknown	Door to door/in person	1	7938.00	4
21	Unknown	Internet	4	6227.50	5
22	Unknown	Direct call	5	6202.11	6
23	Unknown	Not Available	10	131.50	7

===== Query 3 =====

Empty DataFrame

Columns: [Person_ID, Occupation, Age, Age_range, incident_count, total_loss]

Index: []

===== Query 4 =====

	Fraud_kind	cafc_reports	cafc_share_pct	canada_rank	ftc_reports
0	Merchandise	15	12.5	2	383441
1	Investments	14	11.7	3	116071
2	Prize	1	0.8	7	97350
3	Identity Fraud	75	62.5	1	1135291
4	Romance	5	4.2	5	59490
5	Loan	2	1.7	6	23395
6	Job	7	5.8	4	104946
7	Fraudulent Cheque	1	0.8	7	13616

	ftc_share_pct	us_rank	share_difference_pct
0	19.8	2	-7.3
1	6.0	3	5.7

2	5.0	5	-4.2
3	58.7	1	3.8
4	3.1	6	1.1
5	1.2	7	0.5
6	5.4	4	0.4
7	0.7	8	0.1